Introduction to Japan and Japanese Language: Origin, history, Education, Family relations, Work culture and daily life, Travel and tourism, Cultural aspect and Sports. Basic phrases and greetings. Introducing oneself (Jikoshoukai) in Japanese.

Introduction to Hiragana script and its usage Written structure: Scripts- Hiragana, Katakana, Kanji

Spoken structure: Valid sound patterns, Consonants and vowels – Hiragana Script Characters (10) from Aa to Ko: Stroke order writing, practice with flash cards II General words based on completed hiragana characters(10)

Characters (15) from Ga to Zo: Stroke order writing, practice with flash cards General words based on completed hiragana characters(10)

Characters (15) from Ta to No: Stroke order writing, practice with flash cards General words based on completed hiragana characters (15)

Characters (15) from Ha to Po: Stroke order writing, practice with flash cards General words based on completed hiragana characters(15)

Characters (16) from Ma to N: Stroke order writing, practice with flash cards General words based on completed hiragana characters (21)

Rules for sound prolongation and its expression using hiragana. Rules for writing compound characters (Small ‘Ya’,’Yu’,’Yo’). Rules for writing Double consonants, words on compound characters.

III

Counting in Japanese Basic numbers (1 to 10), 2, 3 and 4 digit numbers. Reading and Writing from digits to Japanese and vice versa. Vocabulary- Weekdays (youbi), body parts, Family (Khazoku) in Japanese. Seasons, Time, Months (gatsu) in Japanese.

IV

Grammar Basic sentence pattern ‘A wa B desu’, ‘A wa B desu ka’. Introduction of particles ‘wa’and ‘ka’ Copula Desu/Dewa Arimasen/Deshita/Dewa Arimasen Deshita.

Particle ‘no’ possession marker, Particle ‘Mo’ Daily life phrases.

V

Kara made grammar pattern (From___to__) Kore, Sore, Are-This, that, that over there “Place ni object ga arimasu/ imasu” grammer pattern. Existence of living/ non-living things. Direction marker Particle ‘e’ using grammar pattern “Place ‘e’ ikimasu/ kimasu/ kaerimasu”

5 1. Text Books 2.

Japanese Kana Script Practice Book Volume 1: Hiragana; Patki A. (2020), Daiichi Japanese Language Solutions. ISBN: 9788194562900

Marugoto Starter (A1) Rikai – Course book for communicative Language Competences; The Japan Foundation (2014), Goyal Publishers. ISBN: 9788183078047

E-Books Online 1. 1.

Marugoto Starter (A1) Rikai – Course book for communicative Language Competences; The Japan Foundation, Sanshusha Publishing

https://a1.marugotoweb.jp/en/index.php
https://minato-jf.jp/

Course Code: CE4O002

Course Name: Disaster Management

Course Outcomes:

Students should be able to:

CO1. Acquire the knowledge about disasters and its management.

CO2. Understand the role of different government & social agencies for disaster management.

CO3. Classify the disasters, its preparedness and responses.

CO4. Evaluate the disaster preparedness and extent of risk and cost assessment.

CO5. Solve the crises using the management and different roles involved in it.

Unit	Content	Duration
I	Understanding Natural Disasters: Natural disasters, which have been categorized as hydrological, wind-related, geo-physical, hydro-geological and climatic, discuss the causes and impacts, along with past illustrations and geographical distribution. Flood, Drought, Cyclone, Earthquakes, Landslides, Avalanches, Volcanic Eruptions, Heat and Cold Waves Climate Change: Global Warming, Sea Level Rise, Ozone Depletion.	6
II	Man-Made Disasters: Nuclear Disasters, Chemical Disasters, Biological Disasters, Building Fire, Coal Fire, Forest Fire, Oil Fire, Air Pollution, Water Pollution, Deforestation,IndustrialPollution,RoadAccidents,RailAccidents,AirAccidents, Sea Accidents	6
III	Risk & Cost Assessment: Geographical conditions, Population, Living habits, Threats, Extent of damages to the lives, agricultural area, industrial units, Awareness & Safety Program. Relief arrangement & essential components, Shelters, Rescue & searchtools&equipment,transportfacilities.Costassessmentofeachunitand funding	6
IV	Disaster Preparedness: Concept and Nature, Planning, Disaster Preparedness for People with Special Needs/Vulnerable Groups, Disaster Preparedness with Relevance to Housing, Infrastructure and Livestock, Community Based Disaster Preparedness Plan, Role of Information, Education, Communication, and Training, Disaster Preparedness: Role Play by Government and Social Groups. Information Technology: Role in Disaster Preparedness with Special Reference to Geographical Information System, Use and Application of Emerging Technologies in Disaster Preparedness.	6
V	Disaster Response: Disaster Response Plan, Communication, Participation, and Activation of Emergency Preparedness Plan, Logistics Management, Needs and Damage Assessment, Psychological Response, Trauma and Stress Management. Rumor and Panic Management, Minimum Standards of Relief, Managing Relief, Funding Relief, Recovery	8

VISION	MISSION
To be recognized for excellent engineering, developing global leaders both in educational and research in the domain of computer science and wireless engineering.	1. To create self-learning environment by facilitating leadership qualities, team spirit and ethical responsibilities. 2. To improve department-industry collaboration, interaction with professional society through technical knowledge and internship program. 3. To promote research and development with current techniques through well qualified resources in the area of computer science and wireless engineering.

Program: B. Tech. in Artificial Intelligence

Semester	Course Code	Name of the course	L	T	P	Credits
IV	CS4Oxxx	Open Elective-II (Business Data Processing)	3	0	0	3

Prerequisites for the course	
1.	Basic Computer Literacy.
2.	Fundamental Knowledge of Information Technology.

Prior Reading Material/useful links	
1.	https://www.geeksforgeeks.org/data-science/introduction-to-data-processing/
2.	https://www.simplilearn.com/what-is-data-processing-article
3.	https://www.writerinformation.com/insights/what-is-data-processing-and-what-are-the-processes-involved-

Course Outcomes:

Sr. No	Course Outcome No.	CO statement
1	CO1	Student will be able to understand the concepts of data, information, and various types of data processing systems used in business environments.
2	CO2	Student will be able to describe the structure and functioning of computer systems in data processing.
3	CO3	Student will be able to demonstrate proficiency in creating, formatting, and managing business documents using word processing software.
4	CO4	Student will be able to apply spreadsheet tools to perform calculations, organize data, create charts, and carry out statistical analysis for business decision-making.
5	CO5	Student will be able to design and deliver effective business presentations using presentation software with appropriate use of multimedia and animation tools.

Syllabus:

Course Contents		Hours
Unit I	**Introduction to Business Data Processing:** Data Concept and use of data in computing, Difference between data and information, Role of data in business decision-making, Data Processing Concept and advantages of data processing, Types of data processing: Manual,	09

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	Mechanical, and Electronic, Applications of data processing in business and management, Concepts of Business Data Processing, Basic data processing cycle, Types of Data Processing Systems, Advantages and Limitations of Different Processing Methods.	
Unit II	**Computer Systems for Data Processing:** Overview of Computer System Components in Data Processing, Multiprogramming Systems and Multiprocessing Systems, Distributed Data Processing: Concept, Architecture, and Advantages, Data Storage Concepts: Master File, Transaction File, Intermediate Files, Back-up Files, Data Security and Backup Procedures.	**08**
Unit III	**Word Processing Applications:** Introduction to Word Processing, Applications and Features of Word Processors, Menus and Toolbars in MS Word, Writer, Creating, Entering, Saving, and Printing Documents, Editing and Formatting Text, Using Mail Merge for Bulk Communication, Introduction to Macros in Word Processing.	**07**
Unit IV	**Spreadsheet Applications:** Introduction to Spreadsheet Software, Menus, Toolbars, and Basic Spreadsheet Operations, Preparing Tables and Charts, Data Sorting and Filtering, Using Built-in Functions and Formulas, Statistical Applications in Spreadsheets.	**06**
Unit V	**PowerPoint Presentation Software:** Creating, saving, and editing slides, Using templates, themes, and layouts, Inserting text, images, charts, and multimedia, Applying transitions and animations, Preparing professional business presentations, Running slide shows effectively.	**06**

Text Books	
1.	V. K. Kapoor, Introduction to Computer Data Processing & System Analysis, Sultan Chand & Sons.
2.	Joyce Cox & Joan Lambert, Microsoft Access 2010 Step by Step, Microsoft Press.
3.	Bhadka Harsad and Sharma Priyanka, Business Data Processing, LAP Lambert Academic Publishers.

Reference Books	
1.	Foster Provost & Tom Fawcett, Data Science for Business; O'Reilly Media Publishing House
2.	S.S. Shrivastava, MS Office, Laxmi Publications.

Useful Links	
1.	https://www.youtube.com/watch?v=uvVsyCR4-7c
2.	https://www.youtube.com/watch?v=b3GjKrArLkw

Program: B.Tech. in CSE (DS)

Open Elective Subject for IV Semester CSE (Data Science) Branch Code – CSE(DS)

Semester	Course Code	Name of the course	L	T	P	Credits
IV	4DSO002B	**German Foreign Language**	3	0	0	3

Prerequisites for the course	
1.	NIL

Prior Reading Material/Useful links	
1.	A selection of texts compiled by the German Section of the Department of Foreign Languages
2.	Material compiled by the German Section of the Department of F.L.

Course Outcomes:

Sr. No	Course Outcome	CO statement
1	CO1	Develop awareness of the nature of language and language learning
2	CO2	Encourage positive attitudes towards speakers of other languages and a sympathetic approach to other cultures
3	CO3	Develop the language proficiency required to communicate effectively in German at level basics
4	CO4	Provide enjoyment and intellectual stimulation
5	CO5	Offer insights into the culture and society of countries and communities where German Language is spoken

Syllabus:

Course Contents		Hours
Unit I	Introduction to Germany – its culture and people Pronunciation – BASIC and ADVANCED Basic Greetings and Self-Introduction Grammar- Nouns-genders, article Grammar- Nouns - Plural forms Vocabulary- Months, weekdays and daytimes and number system Vocabulary- Time and date Grammar.	**[6Hrs]**
Unit II	Personal Pronouns Vocabulary-Family, professions Vocabulary- Directions, Common words Vocabulary –Job-Related and Modes Of Transport Grammar r- Irregular verbs Grammar- Modal verbs and Imperative Verbs WH.	**[6Hrs]**
Unit III	Possessive Pronouns Vocabulary- House, Furniture and Draperies Vocabulary- Food and Drinks Grammar-Regular verbs Vocabulary- Vegetables and fruits Grammar.	**[6Hrs]**
Unit IV	Questions Vocabulary- Body parts and Clothes Grammar – Sentences-types and Framing. Grammar-Imperative sentences and Framing questions Vocabulary	**[6Hrs]**
Unit V	Common Places and Hobbies Grammar- Adjectives and Opposites. Test –Viva and Written, Formation and use of language elements and enhancing skills in German Language.	**[6Hrs]**

Text Books	
1.	Brinker-Gabler, Gisela (Hrsg.): Deutsche Dichterinne vom 16.Jahrhundert bis zur Gegenwart,Frankfurt/M. 1986.

2.	Stephan/Venske/Weigel: Frauenliteratur ohne Tradition? Neun Autorinnenporträts, Frankfurt/M. 1987.
3.	Spillmann, Hans.O: Einführung in die germanistische Linguistik. Berlin: Langenscheidt Vlg.2004

Reference Books	
1.	Brinker-Gabler, Gisela (Hrsg.): Deutsche Literatur von Frauen, München 1988. (Zwei Bände) - Brügmann, Margret: Amazonen der Literatur, Amsterdam 1986.
2.	Weigel, Sigrid: Die Stimme der Medusa, 1988.

Useful Links	
1.	https://onlinecourses.nptel.ac.in/noc21_hs30/preview

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3	Prof. Chetan Padole	Asst. Prof.	JDCOEM, Nagpur	
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Course Code	IT
Version	JD/Acd/CSE-DS/2023-27/2
Date and year of release	2026-27

JAIDEV EDUCATION SOCIETY'S

J D COLLEGE OF ENGINEERING AND MANAGEMENT

KATOL ROAD, NAGPUR

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(An Autonomous Institute, with NAAC "A" Grade)

Affiliated to DBATU, RTMNU & MSBTE Mumbai

Department of Computer Science & Engineering

"A Place to Learn, A Chance to Grow"

Session: 2023-27

VISION	MISSION
To be recognized for excellent engineering, developing global leaders both in educational and research in the domain of computer science and wireless engineering.	1. To create self-learning environment by facilitating leadership qualities, team spirit and ethical responsibilities. 2. To improve department-industry collaboration, interaction with professional society through technical knowledge and internship program. 3. To promote research and development with current techniques through well qualified resources in the area of computer science and wireless engineering.

Program: B. Tech. in Computer Science & Engineering

Semester	Course Code	Name of the course	L	T	P	Credits
III	**CS3O001**	**Open Elective-I (Ethical Hacking)**	3	0	0	**3**

Prerequisites for the course	
1.	Basics of Universal Human Value and entrepreneurial behaviour.
2.	Knowledge of the social, ethical and culture and Consumer Affairs

Prior Reading Material/useful links	
1	https://www.google.co.in/books/edition/Entrepreneurship_and_Economic_Developmen/deeFDAAAQBAJ?hl=en&gbpv=1&dq=Innovation+and+Entrepreneurship+Development&printsec=frontcover
2	https://www.google.co.in/books/edition/The_Role_of_Innovation_and_Entrepreneurs/klZYEAAAQBAJ?hl=en&gbpv=1&dq=Innovation+and+Entrepreneurship+Development&printsec=frontcover

Course Outcomes:

Sr. No	Course Outcome number	CO statement
1	CO1	Students will be able to Identify the changing roles, traits of entrepreneurs for the growth of entrepreneurship.
2	CO2	Students will be able to Assess internal and external factors affecting entrepreneurship and apply theories to overcome them.
3	CO3	Students will be able to Analysis of market and Developing business plan process.
4	CO4	Students will be able to develop reports and analyse various funding offered by different financial institutes.
5	CO5	Students will be able to Develop the role and schemes of government to support entrepreneurship

Syllabus:

Course Contents		Hours
Unit I	**Ethical Hacking Overview & Vulnerabilities:** Understanding the importance of security, Concept of ethical hacking and essential Terminologies Threat, Attack, Vulnerabilities, Target of Evaluation, Exploit. Phases involved in hacking.	08
Unit II	**Footprinting & Port Scanning:** Foot printing - Introduction to foot printing, Understanding the information gathering methodology of the	07

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	hackers, Tools used for the reconnaissance phase. Port Scanning - Introduction, using port scanning tools, ping sweeps, Scripting Enumeration-Introduction, Enumerating windows OS & Linux OS	
Unit III	**System Hacking**: Aspect of remote password guessing, Role of eavesdropping ,Various methods of password cracking, Keystroke Loggers, Understanding Sniffers ,Comprehending Active and Passive Sniffing, ARP Spoofing and Redirection, DNS and IP Sniffing, HTTPS Sniffing.	07
Unit IV	**Hacking Web Services & Session Hijacking:** Web application vulnerabilities, application coding errors, SQL injection into Back-end Databases, cross-site scripting, cross-site request forging, authentication bypass, web services and related flaws, protective http headers Understanding Session Hijacking, Phases involved in Session Hijacking, Types of Session Hijacking, Session Hijacking Tools	07
Unit V	**Hacking Wireless Networks:** Introduction to 802.11,Role of WEP, Cracking WEP Keys, Sniffing Traffic, Wireless DOS attacks, WLAN Scanners, WLAN Sniffers, Hacking Tools, Securing Wireless Networks.	07

Text Books	
1.	Kimberly Graves, "Certified Ethical Hacker", Wiley India Pvt Ltd, 2010
2.	Michael T. Simpson, "Hands-on Ethical Hacking & Network Defense", Course Technology, 2010
3.	Rajat Khare, "Network Seuciryt and Ethical Hacking", Luniver Press, 2006
Reference Books	
1.	Ramachandran V, BackTrack 5 Wireless Penetration Testing Beginner's Guide (3rd ed.). Packt Publishing, 2011
2.	Thomas Mathew, "Ethical Hacking", OSB publishers, 2003

Useful Links	
1	https://www.google.co.in/books/edition/Creativity_and_Innovation_in_Entrepreneu/f1c7EAAAQBAJ?hl=en&gbpv=1&dq=Innovation+and+Entrepreneurship+Development&printsec=frontcover
2	https://www.google.com/url?sa=t&rct=j&q=&esrc=s&source=web&cd=&cad=rja&uact=8&ved=2ahUKEwj14Jaw9L__AhWgU2wGHaLcCNUQFnoECCYQAQ&url=https%3A%2F%2Fwww.dpgpolytechnic.com%2Fdownloads%2Ffiles%2Fn5c419bcb66c09.pdf&usg=AOvVaw3cbsLd-So8C6Pgj0LW3S35

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JDCOEM, Nagpur

VISION	MISSION
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VISION	MISSION
"To develop competent and committed Electrical Engineers to serve the society"	1. To impart quality education in the field of Electrical Engineering. 2. To be excellent learning centre through research and industry interaction.

Program: B.Tech in Electrical Engineering

Semester	Course Code	Name of the course	L	T	P	Credits
IV	EE4O002	Green Energy Management	3	0	0	3

Prerequisites for the course	
1	Energy, Environment and Renewable Energy Technologies

Prior Reading Material/useful links	
1	https://archive.nptel.ac.in/courses/112/107/112107209/

Course Outcomes:

Sr. No	Course outcome number	CO statement
1	CO1	Recall the fundamental concepts, scope, and evolution of Green Management and its relevance in the 21st century.
2	CO2	Explain the role of internal and external organizational environments in shaping sustainable management, CSR, and CER practices.
3	CO3	Apply the concepts of ecological economics, sustainability indicators, and life cycle assessment to real-world green management problems.
4	CO4	Analyze environmental standards (ISO 14001, ISO 14064), green financing mechanisms, and their implications for corporate strategy.
5	CO5	Evaluate the effectiveness of green techniques, tax incentives, and eco-commerce models in achieving sustainable business operations.
6	CO6	Design and propose practical, innovative, and sustainable business models integrating green energy management principles.

Syllabus:

Course Contents	
Unit I	The concept of green management; evolution; nature, scope, importance and types; developing a theory; green management in India; relevance in twenty first century. (7 hours)
Unit II	Organizational environment; internal and external environment; Indian corporate structure and environment; how to go green; spreading the concept in organization; Environmental and sustainability issues for the production of high-tech components and materials, life cycle analysis of materials, sustainable production and its role in corporate social responsibility (CSR) and corporate environmental responsibility (CER). (8 hours)
Unit III	Approaches from ecological economics; indicators of sustainability; ecosystem services and their sustainable use; bio-diversity; Indian perspective; alternate theories. (7 hours)
Unit IV	Environmental reporting and ISO 14001; climate change business and ISO 14064; green financing; financial initiative by UNEP; green energy management; green product management (7 hours)
Unit V	Green techniques and methods; Definition; green tax incentives and rebates (to green projects and companies); green project management in action; business redesign; eco-commerce models (7 hours)
Text Books	
1	Green Management: Environmental Initiatives for Sustainable Development – C. S. Rao, Wiley Eastern Ltd.
2	Corporate Environmental Management – Stefan Schaltegger, Roger Burritt, Holger Petersen, Greenleaf Publishing.
Reference Books	
1.	Sustainable Development: Concepts and Practices – M. Karpagam, Sterling Publishers.
2.	Green Business: Principles and Practices – Nevin Cohen & Dustin Mulvaney, Sage Publications.
3.	Environmental Management and Audit – S. K. Bhatia, Atlantic Publishers.
4.	ISO 14001 Environmental Systems Handbook – Ken Whitelaw, Routledge.
Useful links	
1	https://archive.nptel.ac.in/courses/112/107/112107209/

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Secretary BOS Chairman BOS Dean Academics Chairman Acd. Council

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(An Autonomous Institute, with NAAC "A" Grade)
Affiliated to DBATU, RTMNU & MSBTE Mumbai
Department of Electronics and Telecommunication Engineering
"Rectifying Ideas, Amplifying Knowledge"
Session 2024-25

VISION	MISSION
"To be a Department providing high quality & globally competent knowledge of concurrent technologies in the field of Electronics and Telecommunication."	1. To provide quality teaching learning process through well-developed educational environment and dedicated faculties. 2. To produce competent technocrats of high standards satisfying the needs of all stakeholders.

Program: B.Tech in Electronics and Telecommunication Engineering

Semester	Course Code	Name of the course	L	T	P	Credits
IV	ET4O002	Digital Marketing	3	0	0	3

Prerequisites for the course	
1	Basic Knowledge of Social Media.

Prior Reading Material/ useful links	
1	https://www.youtube.com/watch?v=nkNHn0VqVBA&list=PL9ooVrP1hQOH-nMMFeSG-DFwyATtUg4YA

Course Objectives:

Sr.No	Course Objectives
1	To familiarize the students with the concepts and techniques used in digital marketing its applications.
2	To study various digital social media.
3	To explain different techniques used in digital marketing.
4	To enable students to apply different analysis techniques used in digital marketing.

Course Outcomes: At the end of the course, students will be able to:

Sr.No	Course outcome number	CO statement
1	CO1	Understand marketing in digital era and online tools for digital marketing.
2	CO2	Apply different types of Search Engine Optimization techniques.
3	CO3	Analyze different types of marketing techniques like Facebook marketing.
4	CO4	Understand the Social Media Optimization techniques.

Syllabus:

Course Contents	
Unit I	**Introduction to Digital Marketing:** Meaning of Digital Marketing, Differences from Traditional Marketing, Return of Investments on Digital Marketing vs. Traditional Marketing, E Commerce, Tools used for successful marketing, SWOT Analysis of Business for Digital Marketing, Meaning of Blogs, Websites, Portal and Their Differences, Visibility, Visitor Engagement, Conversion Process, Retention, Performance Evaluation. **[6 Hrs]**
Unit II	**Search Engine Optimization (SEO):**On page Optimization Techniques, Off Page Optimization Techniques, Preparing Reports, Creating Search Campaigns, Creating Display Campaigns. **[5 Hrs]**
Unit III	**Social Media Optimization (SMO):**Introduction to Social Media Marketing, Advanced Facebook Marketing. **[6 Hrs]**
Unit IV	**Social Media Marketing:** Word press Blog Creation, Twitter Marketing, LinkedIn Marketing, Instagram Marketing, social media Analytical Tools.Search Engine Marketing:Meaning and Use of Search Engine Marketing, Tools used — Pay Per Click, GoogleAd words, Display Advertising Techniques, Report Generation. **[7Hrs]**
Unit V	**Website Traffic Analysis, Affiliate Marketing and Ad Designing:** Google Analytics, Online Reputation Management, Email Marketing, Affiliate Marketing, Understanding Ad Words Algorithm, Advertisement Designing. **[6 Hrs]**
Text Books	
1	Digital Marketing Strategy: An Integrated Approach to Online Marketing,Simon Kingsnorth, 1st Edition. , Philadelphia, PA : Kogan Page, 2016.
2	How to be a killer Digital Marketing Manager, Tom Bowden & Tom Jepson.
3.	Digital Marketing, Dave Chaffey & Fiona Ellis-Chadwick, 6th Edition, Pearson Publication.
Reference Books	
1	Digital Marketing,Seema Gupta ,3rd Edition, Mcgraw hill
2	Digital Marketing for Beginners : A Road Map to Successful Career in Digital Marketing**,** V Venkata Krishna
3	The Art of Digital Marketing: The Definitive Guide to Creating Strategic, Targeted, and Measurable Online Campaigns,Ian Dodson.
Useful links	
1	https://www.youtube.com/watch?v=ppuiuwNPojU
2	https://www.youtube.com/watch?v=G6DmDqYLWL8

Secretary	Chairman	Chairman
BOS	**BOS**	**Acd. Council**

VISION	MISSION
To be a center of excellence for learning and research in Mechanical Engineering.	1. To provide high quality, innovative and research environment in mechanical engineering. 2. To impart soft skill and hard skill to achieve institutional vision.

Semester	Course Code	Name of the course	L	T	P	Credits
IV	ME4O002	Open Elective II- Industry 4.0	3	0	0	3

Prerequisites for the Course	
1.	Workshop Technology
2.	Basics of Computer Technology

Prior Reading Material/Useful links	
1.	Sabina Jeschke, Christian Brecher, et al., Industrial Internet of Things: Cybermanufacturing Systems, Springer, 2017
2.	https://nptel.ac.in/courses/106105195

Course Objectives:

Sr.No	Statement
1.	To introduce students to the concept and evolution of Industry 4.0.
2.	To understand enabling technologies such as IoT, cyber-physical systems, and smart sensors.
3.	To explore applications of Industry 4.0 in manufacturing and service sectors.
4.	To discuss cybersecurity, connectivity, and digital transformation challenges in industries.

Course Outcomes:

Sr. No	Course Outcome Number	CO statement
1.	CO 1	Explain the evolution and key components of Industry 4.0.
2.	CO 2	Describe Industrial IoT and its integration with industrial processes.
3.	CO 3	Identify major technologies such as AI, Big Data, AR/VR used in Industry 4.0.
4.	CO 4	Discuss cybersecurity and network issues in industrial systems.
5.	CO 5	Illustrate real-world applications of Industry 4.0 through relevant case studies.

Unit No 1: Introduction to Industry 4.0 **[6Hours]**

Overview of Industry 4.0: Evolution from Industrial Revolutions to the Fourth Revolution. Key components: Sensing, actuation, and communication in industries. Globalization and emerging issues in smart manufacturing. Introduction to smart factories and LEAN production systems.

Unit No 2: Industrial IoT and Smart Systems **[6Hours]**

Basics of Industrial IoT: Definition and importance. Industrial processes and integration with IoT. Introduction to Cyber-Physical Systems and Next-Generation Sensors. Role of connectivity and networking in industrial systems.

VISION	MISSION
To be a center of excellence for learning and research in Mechanical Engineering.	1. To provide high quality, innovative and research environment in mechanical engineering. 2. To impart soft skill and hard skill to achieve institutional vision.

Unit No 3: Technologies in Industry 4.0 **[6Hours]**

Introduction to key technologies: Artificial Intelligence, Big Data, and Analytics in industries. Basics of Augmented Reality (AR) and Virtual Reality (VR) in industrial applications. Communication protocols in IIoT (overview of MQTT, OPC UA). Introduction to data management in industrial systems.

Unit No 4: Cybersecurity and Emerging Trends in IIoT **[6Hours]**

Importance of cybersecurity in Industry 4.0.Basics of Fog Computing and its role in IIoT. Overview of Software Defined Networks (SDN) in industrial applications. Security challenges in connected industrial systems.

Unit No 5: Applications of Industry 4.0 and IIoT **[6Hours]**

Application domains: Factories and assembly lines, food industry. Case studies: Milk processing and packaging industries, manufacturing industries. Introduction to inventory management and quality control using IIoT.

Text Books	
1.	Alasdair Gilchrist, Industry 4.0: The Industrial Internet of Things, Apress, 2016
2.	Vijay Kumar, Introduction to Industrial IoT and Industry 4.0, CRC Press, 2021

Reference Books	
1.	Klaus Schwab, The Fourth Industrial Revolution, World Economic Forum, 2016
2.	Dominique Petrie and Edward Crawley, Industry 4.0: Technologies and Applications, Springer, 2019

Useful links	
1.	https://www.edx.org/course/industry-4-0-how-to-revolutionize-your-business
2.	https://www.coursera.org/specializations/digital-manufacturing-design-technology
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